

DAY 1 CASE STUDY: Hospital Patient Record

Case Study Question

A hospital wants to build a Patient Record Management System using MongoDB.

The system should:

- Create a database to store patient information.

```
> use patientDB
< switched to db patientDB
```

- Create a collection to store patient records.

```
> db.createCollection("patients")
< { ok: 1 }
```

- Insert one patient record when a new patient is admitted.

```
> db.patients.insertOne({pid:1,name:"Ravi",age:45,disease:"Diabetes",city:"Chennai"})
< {
  acknowledged: true,
  insertedId: ObjectId('6a10090e58283024458b00c1')
}
```

- Insert multiple patient records when importing historical patient data.

```
> db.patients.insertMany([
  {pid:2,name:"Anita",age:30,disease:"Fever",city:"Madurai"},
  {pid:3,name:"Kumar",age:50,disease:"BP",city:"Trichy"},
  {pid:4,name:"Meena",age:25,disease:"Asthma",city:"Salem"},
  {pid:5,name:"Rahul",age:35,disease:"Diabetes",city:"Coimbatore"},
  {pid:6,name:"Priya",age:40,disease:"Heart",city:"Erode"},
  {pid:7,name:"Suresh",age:55,disease:"BP",city:"Vellore"}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a100a6858283024458b00c2'),
    '1': ObjectId('6a100a6858283024458b00c3'),
    '2': ObjectId('6a100a6858283024458b00c4'),
    '3': ObjectId('6a100a6858283024458b00c5'),
    '4': ObjectId('6a100a6858283024458b00c6'),
    '5': ObjectId('6a100a6858283024458b00c7')
  }
}
```

Handle duplicate patient IDs correctly during bulk insert operations.

```
> db.patients.insertMany([
  { pid: 1, name: "Ravi" },
  { pid: 8, name: "Renu" },
], { ordered: false })
```

Retrieve all patient records for administrative review.

```
> db.patients.find()
```

Retrieve one patient record based on patient ID.

```
> db.patients.find({pid:5})
< {
  _id: ObjectId('6a100a6858283024458b00c5'),
  pid: 5,
  name: 'Rahul',
  age: 35,
  disease: 'Diabetes',
  city: 'Coimbatore'
}
```

Filter patient records based on:

Disease

```
> db.patients.find({ disease: "Diabetes" })
```

City

```
> db.patients.find({ city: "Chennai" })
```

Age range

```
> db.patients.find({ age: { $gte: 30, $lte: 50 } })
```

Display only selected fields (such as name and disease) for doctors.

```
> db.patients.find({}, { name: 1, disease: 1, _id: 0 })
```

Remove the collection or database when the hospital migrates to a new system.

```
> db.patients.drop()
< true
```

```
> db.dropDatabase()
< { ok: 1, dropped: 'patientDB' }
```

Sample Data (Patients Collection)

pid Int32	name String	age Int32	disease String	city String
1	"Ravi"	45	"Diabetes"	"Chennai"
2	"Anita"	30	"Fever"	"Madurai"
3	"Kumar"	50	"BP"	"Trichy"
4	"Meena"	25	"Asthma"	"Salem"
5	"Rahul"	35	"Diabetes"	"Coimbatore"
6	"Priya"	40	"Heart"	"Erode"
7	"Suresh"	55	"BP"	"Vellore"